Unit: Unit 2: Exponents Radicals
Topic: Operations with Radicals

Name: ________________
Date: __________

Foundation (Jalapeno)

Q1. Simplify $\sqrt{9} + \sqrt{16}$
(A) $3 + 4$
(B) $5 + 2$
(C) 7
(D) 1
(E) 25

Q2. Simplify $\sqrt{4} \cdot \sqrt{9}$
(A) 6
(B) 12
(C) $2 \cdot 3$
(D) 3
(E) 1

Q3. In a basketball game, the home team scores $\sqrt{256}$ points and the visiting team scores $\sqrt{144}$ points. What is the difference in their scores?
(A) 8
(B) $16 - 12$
(C) 4
(D) 20
(E) 112

Q4. A sprinter runs $\sqrt{36}$ meters in the first heat and $\sqrt{49}$ meters in the second heat. What is the total distance she runs?
(A) 10
(B) $6 + 7$
(C) 5
(D) 13
(E) 25

Q5. Simplify $\frac{\sqrt{16}}{\sqrt{4}}$
(A) 2
(B) 1
(C) 4
(D) 8
(E) 16

Q6. A soccer player kicks the ball $\sqrt{25}$ yards and it goes over the head of a player who is $\sqrt{9}$ yards away. What is the difference between the distance the ball was kicked and the player's position?
(A) 6
(B) 4
(C) 16
(D) $5 - 3$
(E) 2

Q7. In a long jump competition, a jumper lands $\sqrt{64}$ meters away from the launch point. What is the distance the jumper covered?
(A) 1
(B) 4
(C) 8
(D) 12
(E) 16

Q8. A tennis player serves the ball $\sqrt{100}$ kilometers per hour and her opponent returns it $\sqrt{81}$ kilometers per hour. What is the sum of their speeds?
(A) 20
(B) $10 + 9$
(C) 16
(D) 30
(E) 45

Open Ended Questions (FRQ)

Q9. In a high jump competition, the bar is set at a height of $\sqrt{196}$ centimeters and a jumper clears it by $\sqrt{49}$ centimeters. Calculate the jumper's total height.

Q10. A cyclist covers $\sqrt{400}$ meters in 2 minutes and $\sqrt{225}$ meters in 1.5 minutes. Calculate the cyclist's average speed.

Chef's Tips

- Tip 1: Ensure students use the correct radical notation.
- Tip 2: Pay attention to units and conversion where necessary.
- Tip 3: Remind students to simplify their answers.